

20260629EB_MN (ATP, NCT, Protein, IF)

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2026年6月29日 星期一

This batch has A LOT of non-MNs.

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 6 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2026年7月1日 星期三

T2 -> T3

2026年7月3日 星期五

T3 -> T4 (~14 mL for over-weekend)

2026年7月6日 星期一

T4 -> T5

2026年7月8日 星期三

T5 -> T6

2026年7月10日 星期五

T6 -> T6 (~14mL for over-weekend)

2026年7月13日 星期一

T6

2026年7月14日 星期二

T6 -> T7

Dissociation

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 3mL 0.25% trypsin for each (prepare 8mL 0.25% trypsin with 220U DNaseI)
3. Incubate the tubes in 37C.
4. pipette*10 at 5min, 10min.
5. Stop reaction by adding 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
6. add 8mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.

8. centrifuge at 200xg, 4min

cell info ^

	A	B	C	D
1		cell	passage	well number
2	A	WT11	P19+5	6
3	B	WT11	P19+5	6

Table1 ^

	A	B
1		volume (mL)
2	trypsin	4
3	HI-FBS/CTWM	8

Table3 ✓

	A	B	C	D	E	F	G	H	I	J	K
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	A	2.75	100	5	13.75	24I (A) 92	24	300	10	1091	
3	B	3.34	98	5	16.7	24I (B) 92	24	300	10	898	
4						24II (A) 110	24	300	6	655	
5						24II (B) 110	24	300	6	539	
6						24II (A) 111	24	300	4	436	
7						24II (B) 111	24	300	4	359	
8						24II (A) 185	24	300	2	218	
9						24II (B) 185	24	300	2	180	I accidentally plated fewer cells for these two wells
10						6I (A)	6	1638	4		the rest
11						6II (B)	6	1583	6		the rest

24I ^

	1	2	3	4	5	6
A	92 A (1uL/well)					A no virus
B						A no virus
C	92 B (1uL/well)					B no virus
D						B no virus

24II

	1	2	3	4	5	6
A	110 A (1uL/well)					
B	111 A (1.5uL/well)				185 A (3uL/well)	
C	110 B (1uL/well)					
D	111 B (1.5uL/well)				185 B (3uL/well)	

6I (A) ^

	1	2	3
A			
B			

6II (B) ^

	1	2	3
A			
B			

- 2026年7月16日 星期四
- T7 -> T8
- 2026年7月17日 星期五
- 1/2 T8 -> 1/2 T8
- 2026年7月20日 星期一
- 1/2 T8 -> 1/2 T8
- 2026年7月22日 星期三
- 1/2 T8 -> 1/2 T8

NCT imaging with 2mM MMS (DMSO, olaparib)

**2mM MMS did not slow down import rate this time...

time table for NCT treatment_MMS_Plate II-BatchA											
	A	B	C	D	E	F	G	H	I	J	K
1	NCT (plate I)										
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging		
3											
4	A	A1	MMS 0mM	DMSO	2.5mM gluc	1mM	1	10:08	11:11		
5	A'	A2	MMS 2mM	DMSO	2.5mM gluc	1mM	3	11:37	12:35	the import looked fine this time. Saw a lot of 640-labeled neurites (a bit weird). I checked the primary dilution and it was less than 100ul. Unless I forgot to add MMS itself (I don't think so?)	7/23/2026: Cells died, so I did add the MMS.
6	B	A3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	5	12:58	13:56		
7	B'	A4	MMS 2mM	olaparib 200nM	2.5mM gluc	1mM	7	14:25			
8											
9	A	B1	MMS 0mM	DMSO	2.5mM gluc	1mM	2	10:55	11:52		
10	A'	B2	MMS 2mM	DMSO	2.5mM gluc	1mM	4	12:15	13:18		
11	B	B3	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	6	13:34	14:43	delayed > 5min	
12	B'	B4	MMS 2mM	olaparib 200nM	2.5mM gluc	1mM	8	15:03	16:01		

fileID_treatment_NCT_20260722_MN_110,111						
	A	B	C	D	E	F
1	image	reporter	treatment	drug	time	pyruvate
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	DMSO	1h	yes
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	DMSO	1h	yes
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	2mM MMS	DMSO	1h	yes
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	2mM MMS	DMSO	1h	yes
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0MM MMS	200nM olaparib	1h	yes
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	0mM MMS	200nM olaparib	1h	yes
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	2mM MMS	200nM olaparib	1h	yes
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	2mM MMS	200nM olaparib	1h	yes
10	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	DMSO	1h	yes
11	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	DMSO	1h	yes
12	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	2mM MMS	DMSO	1h	yes
13	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	2mM MMS	DMSO	1h	yes
14	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0MM MMS	200nM olaparib	1h	yes
15	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	0mM MMS	200nM olaparib	1h	yes
16	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	2mM MMS	200nM olaparib	1h	yes
17	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	2mM MMS	200nM olaparib	1h	yes

Perceval: MMS with titration

drug calculation_MMS																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
1														batchA		batchB		
2			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution (mM)	working (mM)	dilution factor from 1st dil.	total volume (mL)	volume of 1st dil.	volume of media	treatment time	image time	treatment	image time	
3	MM S	1							0		1000	0	1000	15:41	16:41	18:37	19:37	
4		2	11800	100	100	1	99	118	1	118	1000	8.5	1000	16:00	17:01	19:05	20:03	
5		3	11800	100	100	1	99	118	1.5	78.6666666667	1000	12.9	1000	16:28	17:27	19:23	20:25	
6		4	11800	100	100	1	99	118	2	59	1000	17.2	1000	16:48	17:47	19:41	20:43	
7		5	11800	100	100	1	99	118	3	39.3333333333	500	13.0	500	17:07	18:06	20:10		
8		6	11800	100	100	1	99	118	5	23.6	1000	44.2	1000	17:33	18:33	20:30		
9										total	5500	96.0						

fileid_treat_20260709_MN_92						
	A	B	C	D	E	F
1	image	treatment	drug	treat_time	pyruvate	replicate
2	vLH92_MN_60x-1xzoom_0000	0mM MMS	NT	1h	yes	A
3	vLH92_MN_60x-1xzoom_0001	1mM MMS	NT	1h	yes	A
4	vLH92_MN_60x-1xzoom_0002	1.5mM MMS	NT	1h	yes	A
5	vLH92_MN_60x-1xzoom_0003	2mM MMS	NT	1h	yes	A
6	vLH92_MN_60x-1xzoom_0004	3mM MMS	NT	1h	yes	A
7	vLH92_MN_60x-1xzoom_0005	5mM MMS	NT	1h	yes	A
8	vLH92_MN_60x-1xzoom_0006	0mM MMS	NT	1h	yes	B
9	vLH92_MN_60x-1xzoom_0007	1mM MMS	NT	1h	yes	B
10	vLH92_MN_60x-1xzoom_0008	1.5mM MMS	NT	1h	yes	B
11	vLH92_MN_60x-1xzoom_0009	2mM MMS	NT	1h	yes	B
12	vLH92_MN_60x-1xzoom_0010	3mM MMS	NT	1h	yes	B
13	vLH92_MN_60x-1xzoom_0011	5mM MMS	NT	1h	yes	B

2026年7月23日 星期四

time table for NCT treatment_MMS_Plate II-BatchB										
	A	B	C	D	E	F	G	H	I	J
1	NCT (plate I)									
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	
3										
4	A	A1	MMS 0mM	DMSO	2.5mM gluc	1mM	1	12:15	13:13	
5	A'	A2	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	3	13:35	14:33	
6	B	A3	MMS 2mM	DMSO	2.5mM gluc	1mM	5	14:57	15:55	import looked very normal again...
7	B'	A4	MMS 2mM	olaparib 200nM	2.5mM gluc	1mM	7	16:19	17:17	
8										
9	A	B1	MMS 0mM	DMSO	2.5mM gluc	1mM	2	12:53	13:51	
10	A'	B2	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	4	14:09	15:13	
11	B	B3	MMS 2mM	DMSO	2.5mM gluc	1mM	6	15:32	16:32	
12	B'	B4	MMS 2mM	olaparib 200nM	2.5mM gluc	1mM	8	16:52	17:50	XY01 and XY02 have shifted focus

fileID_treatment_NCT_20260723_MN_110,111						
	A	B	C	D	E	F
1	image	reporter	treatment	drug	time	pyruvate
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	DMSO	1h	yes
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	DMSO	1h	yes
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	0MM MMS	200nM olaparib	1h	yes
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0mM MMS	200nM olaparib	1h	yes
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	2mM MMS	DMSO	1h	yes
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	2mM MMS	DMSO	1h	yes
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	2mM MMS	200nM olaparib	1h	yes
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	2mM MMS	200nM olaparib	1h	yes
10	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	DMSO	1h	yes
11	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	DMSO	1h	yes
12	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	0MM MMS	200nM olaparib	1h	yes
13	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0mM MMS	200nM olaparib	1h	yes
14	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	2mM MMS	DMSO	1h	yes
15	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	2mM MMS	DMSO	1h	yes
16	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	2mM MMS	200nM olaparib	1h	yes
17	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	2mM MMS	200nM olaparib	1h	yes

drop images_20260723_MN_110,111	
	A
1	Label
2	com_111_MN_no-glu_50%_1s_30s_60x_0006_XY1
3	com_111_MN_no-glu_50%_1s_30s_60x_0006_XY2

Perceval: MMS with olaparib

time table for Perceval_MMS_Plate I										
	A	B	C	D	E	F	G	H	I	J
1	Perceval									
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	2.5h imaging
3	batch A									
4	A	A4	MMS 0mM	DMSO	2.5mM gluc	1mM	1	17:23	18:20	
5	B	B4	MMS 0mM	200nM olaparib	2.5mM gluc	1mM	2	17:52	18:50	
6	A'	C4	MMS 2mM	DMSO	2.5mM gluc	1mM	3	18:09	19:10	20:41
7	B'	D4	MMS 2mM	200nM olaparib	2.5mM gluc	1mM	4	18:32	19:33	
8	batch B									
9	A	A4	MMS 0mM	DMSO	2.5mM gluc	1mM	1	18:53	19:51	
10	B	B4	MMS 0mM	200nM olaparib	2.5mM gluc	1mM	2	19:15	20:13	
11	A'	C4	MMS 2mM	DMSO	2.5mM gluc	1mM	3	19:35	20:33	
12	B'	D4	MMS 2mM	200nM olaparib	2.5mM gluc	1mM	4	19:56	20:56	

fileid_treat_20260723_MN_92						
	A	B	C	D	E	F
1	image	treatment	drug	treat_time	pyruvate	replicate
2	vLH92_MN_60x-1xzoom_0000	0mM MMS	DMSO	1h	yes	A
3	vLH92_MN_60x-1xzoom_0001	0mM MMS	200nM olaparib	1h	yes	A
4	vLH92_MN_60x-1xzoom_0002	2mM MMS	DMSO	1h	yes	A
5	vLH92_MN_60x-1xzoom_0003	2mM MMS	200nM olaparib	1h	yes	A
6	vLH92_MN_60x-1xzoom_0004	0mM MMS	DMSO	1h	yes	B
7	vLH92_MN_60x-1xzoom_0005	0mM MMS	200nM olaparib	1h	yes	B
8	vLH92_MN_60x-1xzoom_0006	2mM MMS	DMSO	1h	yes	B
9	vLH92_MN_60x-1xzoom_0007	2mM MMS	DMSO	2.5h	yes	A
10	vLH92_MN_60x-1xzoom_0008	2mM MMS	200nM olaparib	1h	yes	B
11	vLH92_MN_60x-1xzoom_0009	2mM MMS	200nM olaparib	2.5h	yes	A

time table for NCT treatment_hypoglycemia_Plate II-BatchA

	A	B	C	D	E	F	G	H
1	NCT (plate I)							
2	condition#		condition	pyruvate	imaging order	time	imaging	
3	110							
4	A	A5	2.5mM gluc	1mM	1	15:20	16:18	
5	B	A6	0.15mM gluc	1mM	3	16:40		
6	A'	C5	0.15mM gluc	1mM	5	17:56		
7	B'	C6	2.5mM gluc	1mM	7	19:15	20:13	
8	185							
9	A	B5	2.5mM gluc	1mM	2	16:00		
10	B	B6	0.15mM gluc	1mM	4	17:20		
11	A'	D5	0.15mM gluc	1mM	6	18:38	19:36	XY01 shifted focus
12	B'	D6	2.5mM gluc	1mM	8	19:58	20:56	XY01 shifted focus

drop images_20260727_MN_110,185

	A
1	Label
2	com_185_MN_no-glu_50%_1s_30s_60x_0004_XY1
3	com_185_MN_no-glu_50%_1s_30s_60x_0006_XY1

2026年7月29日 星期三

Protein

Batch A

	1	2	3
A	2.5mM gluc		
B	0.15mM gluc		

Batch B			
	1	2	3
A	2.5mM gluc		
B	0.15mM gluc		

^

(Record on 8/27/2026)

**For batch B A1 and B1 (or A3 and B3, I forgot to record it), I scraped the cells a little PBS and add RIPA directly. After adding RIPA, the solution became very viscous.

1. Wash cells with 1mL media and treat them at 16:22
2. Precool centrifuger and eppendorf
3. Prepare RIPA buffer with 2%SDS (250uL RIPA + 25uL 20%SDS)
4. Incubate cells in 37C incubator
5. At 17:22, placed the whole plate on ice. And process the following steps one well by one well.
 - a. remove media carefully with P1000
 - b. took 500uL cold PBS and flush the cells down quickly and strongly to detach the cells
 - c. Collect cells suspension into the pre-cooled eppendorf
 - d. After process every cells, I rinse the well again with 500uL cold PBS
 - e. ** I checked the plate under microscope and most cells were detached. Some cells were still there, but I saw many non-MNs are still attached on the well!! It can be a good way to get rid of non-MNs.
6. centrifuge the cells with 500xg, 2min
7. Add 2.5 µl PMSF solution, 2.5 µl sodium orthovanadate solution and 2.5 µl protease inhibitor cocktail solution (1:100) freshly
8. remove supernatant with P1000 and leave a bit PBS to avoid aspirate out the cell pellet
9. estimate the amount of the left liquid and add approximately same amount of RIPA. I added 40uL this time to not dilute protein too much. Pipette the cells a bit.
10. Boil the lysate at 95C for 5-10min
11. Store the protein in -80C freezer

IF

I did not find what I treated for IF. I think I treated higher concentration of the MMS to see if I can induce stress granules (perhaps 3mM vs. 5mM or control vs. 5mM)

1. fix with 4%PFA for 15min
2. wash with PBS
3. 0.2% triton for 10min
4. 10%FBS blocking
5. 1st Ab o/n

24I IF treatmet							^
	1	2	3	4	5	6	
A							
B							
C							
D							

antibody													/
	A	B	C	D	E	F	G	H	I	J	K	L	
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume	
2	A	C.2	G3BP1	m	500	720	1.44	488	B	1000	720	0.72	
3	A	102	TIA1	Rb	500	720	1.44	594	O	1000	720	0.72	

2026年7月30日 星期四

- 1. wash with PBS*3
- 2. 2nd Ab
- 3. wash with PBS*3
- 4. DAPI
- 5. mounting

2026年8月28日 星期五

I do not know what the conditions are, so I aim to check the antibody and if there is any stress granules formed. According to the phenotype, the treatment might be control vs. 5mM (or 3mM) MMS. I think it's 5mM.

I don't think I add Halo dye since the 640 signal looks mostly identical to 561 (have some puncta that don't exist in other channels and doesn't have nuclear signal.)

G3BP1 has many nuclear staining, but TIA1 does not have nuclear staining.

TIA1 has a few very bright puncta in stress granules which colocalized with G3BP1, but TIA1 have much lower background staining.

Many G3BP1 "puncta" doesn't not have TIA1 puncta.

**I still took some 640 images, but do not use them. I think they are just bleedthrough.

Table2 ^

	A	B	C	D
1		G3BP1	TIA1	note
2	A6	some cells have G3BP1 puncta, but not TIA1	very few cells have TIA1 puncta, but they are all G3BP1+	
3	B6	cannot find G3BP1+/TIA1- puncta	more TIA1+ puncta. Most of them are G3BP1+ (but not very bright)	I guess this group is 5mM MMS
4	C6	some cells have G3BP1 puncta, but not TIA1	very diffused	did not take images since I didn't find many cells
5	D6		more cells with TIA1+ puncta!	